

ENAE644 Term Project

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Abstract—Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaleam animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

I. INTRODUCTION

The goal of this project is to implement and evaluate algorithms from two complementary sub-fields of adversarial motion planning. The first is *deceptive motion planning*, which is concerned with generating trajectories that conceal the agent's true objective from an observer [1], [2]. The second is *goal identification and interception*, which seeks to infer an observed agent's hidden objective and plan a trajectory to intercept it [3], [4]. Rather than studying these in isolation, the project will pit a deceptive planner against an interception-focused planner in a shared environment, providing a direct adversarial evaluation of each algorithm's strengths and limitations. Both agents will employ planning methods at least as sophisticated as RRT*, incorporating learning-based and game-theoretic components on top of their respective base planners.

II. MOTIVATION

Deceptive motion planning has direct applications in video game AI, where non-player characters must exhibit unpredictable yet purposeful behavior [3], [4]. However, the relevance of this problem extends well beyond entertainment into real-world domains such as strategic logistics, privacy-preserving navigation, and military operations. The ability to move deceptively can be the difference between mission success and failure [1], [2], while the complementary problem of identifying and intercepting deceptive agents is equally critical: security and defense systems must be capable of

recognizing when an observed agent is behaving deceptively and responding accordingly [3], [5]. Studying the interplay between deception and counter-deception will provide insight into the robustness of each approach and reveal failure modes that would not surface when evaluating either in isolation.

It is important to note that prior work has explored this coupling in the context of autonomous vehicles [6] and multi-agent navigation [7], [8], but an adversarial pitting in a strategy-game context remains underexplored.

III. RELATED WORK

A. Deceptive Motion Planning

A central challenge in deceptive motion planning is balancing path cost against unpredictability. Nichols et al. [1] address this trade-off with Adversarial RRT*, a sampling-based planner that incorporates a learned deception cost into the RRT* objective. Their approach uses a recurrent neural network as a surrogate observer and penalizes trajectories that the network can classify correctly, achieving low observer accuracy while keeping path length near-optimal. Xu et al. [2] take a complementary approach grounded in probabilistic goal recognition, formalizing deceptive path planning as a mixed-integer program that maximizes goal ambiguity with respect to an observer's belief distribution. Both methods assume a passive observer and evaluate deception in isolation rather than against an active adversary, which is the gap this project aims to address.

B. Goal Recognition and Interception

On the opposing side of the problem, several works focus on inferring an agent's hidden objective and acting on that inference. Xu and Yin [5] introduce *relative goal uncertainty*, an entropy-based metric that quantifies the goal-related information contained in each action, and show how it can be used to control the goal identification process from either side of the interaction. Zeng and Xu [3] extend goal recognition to explicitly account for deceptive agents by using inverse reinforcement learning to recover a behavioral model from the deceiver's historical data, then constructing a game-theoretic interference strategy. In the domain of first-person shooter games, Tastan et al. [4] learn player-specific motion models via inverse reinforcement learning and use particle-filter-based prediction to intercept opponents on partially occluded maps. These works provide the algorithmic foundations for the inter-

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ceptor agent in this project, particularly the combination of learned behavioral models with online prediction.

C. Game-Theoretic and Interactive Planning

A related body of work frames multi-agent planning as a dynamic game in which each agent's strategy depends on the others'. Le Cleac'h et al. [8] propose LUCIDGames, which pairs an unscented Kalman filter for online estimation of other agents' cost functions with a receding-horizon game-theoretic planner, demonstrating real-time performance in autonomous driving scenarios. Espinoza et al. [6] tightly couple prediction and planning through a game-theoretic MPC that uses a multi-agent neural network policy as its predictive model, producing interactive behaviors between an autonomous vehicle and surrounding traffic. While both of these works operate in the autonomous driving domain, they demonstrate the value of jointly reasoning about one's own plan and other agents' likely responses, directly informing the adversarial evaluation framework in this project.

D. Adversary-Aware Motion Planning

Most closely related to the present work, Netter and Vamvoudakis [7] propose a motion planning framework in which a player agent navigates a multi-agent environment while simultaneously identifying and avoiding potential adversaries using Gaussian process classification. Their method includes real-time replanning to avoid likely adversarial agents and distinguishes adversaries from benign agents to prevent unnecessary evasive maneuvers. This project builds on a similar premise but inverts the emphasis: rather than treating interception avoidance as a byproduct of adversary classification, the deceptive agent here will actively minimize observer accuracy as a first-class planning objective, while the interceptor will be a fully autonomous adversary rather than a fixed behavioral model.

IV. FORMAL PROBLEM DEFINITION

This section formalizes the adversarial motion planning problem as a two-agent scenario in which a deceptive agent attempts to reach a hidden goal while concealing its intent, and an interceptor agent seeks to infer the hidden goal and intercept the deceptive agent before the goal is reached.

A. Workspace and Agent Models

We consider a bounded, continuous two-dimensional workspace $\mathcal{W} \subset \mathbb{R}^2$ containing a static obstacle set $\mathcal{O} \subset \mathcal{W}$. The collision-free workspace is defined as $\mathcal{W}_{\text{free}} = \mathcal{W} \setminus \mathcal{O}$. Two agents operate within this workspace: a deceptive agent (Agent D) and an interceptor agent (Agent I).

1) *Deceptive Agent (Agent D)*: The deceptive agent's state at time t is its position $x_D(t) \in \mathcal{W}_{\text{free}}$. Agent D has a finite set of candidate goals $\mathcal{G}_D = \{g_1, g_2, \dots, g_M\} \subset \mathcal{W}_{\text{free}}$ known to both agents, and a true goal $g^* \in \mathcal{G}_D$ that is hidden from Agent

I. The agent executes a trajectory $\xi_D : [0, T] \rightarrow \mathcal{W}_{\text{free}}$ satisfying the boundary conditions $\xi_D(0) = x_{D,0}$ and $\xi_D(T) = g^*$, where $x_{D,0}$ is the initial position and T is the time horizon. The trajectory is subject to kinodynamic constraints including velocity bounds and acceleration limits, which we denote generically as $\xi_D \in \Xi_D$, where Ξ_D is the set of kinodynamically feasible trajectories.

2) *Interceptor Agent (Agent I)*: The interceptor agent's state at time t is its position $x_I(t) \in \mathcal{W}_{\text{free}}$. Agent I starts at position $x_{I,0}$ and executes a trajectory $\xi_I : [0, T] \rightarrow \mathcal{W}_{\text{free}}$ with $\xi_I(0) = x_{I,0}$. The objective of Agent I is to infer the true goal g^* from observations of Agent D's partial trajectory and plan an interception trajectory to reach the predicted goal before Agent D. Agent I's trajectory is also subject to kinodynamic constraints $\xi_I \in \Xi_I$, which may differ from those of Agent D.

3) *Observer Model*: Agent I maintains a belief distribution over the candidate goals, represented as $b_t : \mathcal{G}_D \rightarrow [0, 1]$ where $\sum_{g \in \mathcal{G}_D} b_t(g) = 1$. This belief is updated incrementally based on observations of the partial trajectory $\xi_D[0, t]$. The belief update mechanism will be specified in detail in the following subsections.

B. Deceptive Motion Planning Problem

The deceptive agent seeks to generate a trajectory ξ_D^* that satisfies three competing objectives: (1) reach the true goal g^* , (2) minimize path cost, and (3) maximize the observer's uncertainty about g^* . This multi-objective optimization problem is formalized as

$$\xi_D^* = \arg \min_{\xi_D \in \Xi_D} [\alpha \cdot J_{\text{path}}(\xi_D) + (1 - \alpha) \cdot J_{\text{deception}}(\xi_D)] \quad (1)$$

where $J_{\text{path}}(\xi_D)$ is the path cost, $J_{\text{deception}}(\xi_D)$ quantifies the "revealingness" of the trajectory (i.e., how easily an observer can infer the true goal), and $\alpha \in [0, 1]$ is a weighting parameter that balances path optimality against deception.

The path cost is typically defined as the total path length or control effort:

$$J_{\text{path}}(\xi_D) = \int_0^T \|\dot{\xi}_D(t)\| dt \quad (2)$$

The deception cost is more subtle and can be formulated in multiple ways. Following the Adversarial RRT* framework [1], we employ a learned classifier f_θ (specifically, a recurrent neural network) as a surrogate observer. The classifier maps partial trajectories to probability distributions over goals: $f_\theta : \xi_D \mapsto \Delta(\mathcal{G}_D)$, where $\Delta(\mathcal{G}_D)$ denotes the probability simplex over $\mathcal{G}_D$. The deception cost can then be defined as the negative entropy of the classifier's output:

$$J_{\text{deception}}(\xi_D) = -H(f_\theta(\xi_D)) \quad (3)$$

where $H(p) = -\sum_{g \in \mathcal{G}_D} p(g) \log p(g)$ is the Shannon entropy. High entropy corresponds to high uncertainty (good deception), so minimizing $-H$ is equivalent to maximizing

uncertainty. Alternatively, the deception cost can be defined directly as the classifier's accuracy:

$$J_{\text{deception}}(\xi_D) = P_{\theta}(\text{correct goal} \mid \xi_D) = f_{\theta}(\xi_D)(g^*) \quad (4)$$

An alternative formulation based on probabilistic goal recognition [2], [5] defines the deception cost as the negative goal ambiguity:

$$J_{\text{deception}}(\xi_D) = - \sum_{g \in \mathcal{G}_D} P(g \mid \xi_D) \log P(g \mid \xi_D) \quad (5)$$

where the posterior probability $P(g \mid \xi_D)$ is computed using a probabilistic goal recognition framework that models the likelihood of trajectories under different goal hypotheses.

The optimization problem is subject to the following constraints:

- Collision avoidance: $\xi_D(t) \in \mathcal{W}_{\text{free}}$ for all $t \in [0, T]$
- Kinodynamic feasibility: $\xi_D \in \Xi_D$ (velocity and acceleration limits)
- Boundary conditions: $\xi_D(0) = x_{D,0}$ and $\xi_D(T) = g^*$

1) *Solution Method*: We employ Adversarial RRT* [1], a sampling-based motion planner that extends the RRT* algorithm by incorporating the deception cost $J_{\text{deception}}$ into the cost function used for tree rewiring and path selection. The planner iteratively samples configurations, extends the search tree, and evaluates candidate paths using the combined cost $\alpha J_{\text{path}} + (1 - \alpha) J_{\text{deception}}$, producing trajectories that balance path optimality with unpredictability.

C. Goal Inference and Interception Problem

The interceptor agent must solve two interrelated subproblems:

- 1) Infer the deceptive agent's true goal g^* from observations of the partial trajectory $\xi_D[0, t]$
- 2) Plan an interception trajectory to reach the predicted goal before Agent D.

1) *Goal Inference Subproblem*: The goal inference problem is addressed through a combination of offline learning and online belief updates.

a) *Inverse Reinforcement Learning (IRL) Phase*: Prior to runtime, Agent I uses inverse reinforcement learning to recover a behavioral model of Agent D from historical trajectory data. The IRL problem seeks to recover a reward (or cost) function $R_D(s, a)$ such that the observed trajectories are approximately optimal under this reward function. Following [3], we model the deceptive agent's behavior by learning a reward function that captures both path efficiency and deception, enabling Agent I to predict likely future actions given a goal hypothesis.

b) *Online Belief Update*: During execution, Agent I maintains and updates the belief distribution $b_t(g)$ over candidate goals using Bayes' rule:

$$b_{t+1}(g) \propto P(\xi_D[t, t+1] \mid g, \xi_D[0, t]) \cdot b_t(g) \quad (6)$$

where $P(\xi_D[t, t+1] \mid g, \xi_D[0, t])$ is the likelihood of the observed trajectory segment given goal hypothesis g and

the trajectory history. This likelihood is computed using the behavioral model learned via IRL.

c) *Particle Filter Implementation*: Following [4], we employ a particle filter to maintain and propagate the belief distribution. Each particle represents a hypothesis about the true goal and the agent's current state. The particle filter alternates between a prediction step (using the learned motion model to propagate particles forward) and an update step (reweighting particles based on observed trajectory segments).

The predicted goal at time t is obtained by selecting the maximum a posteriori (MAP) estimate:

$$\hat{g}(t) = \arg \max_{g \in \mathcal{G}_D} b_t(g) \quad (7)$$

2) *Interception Planning Subproblem*: Given the predicted goal $\hat{g}(t)$, Agent I must plan a trajectory to intercept Agent D. The interception planning problem is formulated as:

$$\xi_I^* = \arg \min_{\xi_I \in \Xi_I} T_{\text{intercept}} \quad (8)$$

subject to the constraint that Agent I reaches the predicted goal $\hat{g}$ before Agent D. However, because the prediction $\hat{g}(t)$ evolves over time as new observations are incorporated, a static plan is insufficient.

a) *Game-Theoretic Model Predictive Control (MPC)*: Following [8] and [6], we employ a receding-horizon game-theoretic planner. At each time step t , Agent I solves the finite-horizon optimization problem:

$$\min_{\xi_I[t, t+H]} J_I(\xi_I, \hat{\xi}_D) \quad (9)$$

where H is the prediction horizon and $\hat{\xi}_D$ is the predicted future trajectory of Agent D, computed by forward-simulating the learned behavioral model conditioned on the current belief distribution. The cost function J_I is defined as:

$$J_I(\xi_I, \hat{\xi}_D) = \|\xi_I(T_{\text{intercept}}) - \hat{g}\|^2 + \int_t^{t+H} \|\dot{\xi}_I(\tau)\|^2 d\tau \quad (10)$$

where the first term penalizes distance to the predicted goal and the second term penalizes control effort. The solution yields a planned trajectory for the next H time steps, of which only the first segment is executed before replanning at the next time step.

b) *Replanning*: The MPC framework enables real-time replanning. As Agent I observes new segments of Agent D's trajectory, the belief distribution $b_t(g)$ is updated, the predicted goal $\hat{g}(t)$ may change, and the interception plan is recomputed accordingly.

D. Adversarial Game Formulation

The interaction between the deceptive agent and the interceptor can be formalized as a two-player dynamic game with asymmetric information.

1) Players and Objectives:

- Player 1 (Agent D): Minimizes $J_D = \alpha J_{\text{path}} + (1 - \alpha) J_{\text{deception}}$
- Player 2 (Agent I): Minimizes J_I (interception cost, including time-to-intercept and control effort)

2) *Information Structure*: This is an asymmetric information game. Agent D has complete information: it knows the true goal g^* , the candidate goal set $\mathcal{G}_D$, and can observe Agent I's current state $x_I(t)$. In contrast, Agent I has incomplete information: it knows the candidate goal set $\mathcal{G}_D$ and can observe the partial trajectory $\xi_D[0, t]$, but does not know the true goal g^* . Agent I must infer g^* from observations.

3) *Solution Concept*: Due to the information asymmetry and the sequential nature of the interaction, this problem does not admit a standard Nash equilibrium solution. Instead, it is more appropriately modeled as a Stackelberg game (leader-follower game), in which Agent D (the leader) plans its trajectory anticipating that Agent I (the follower) will respond optimally given its observations. However, Agent D must account for the fact that Agent I does not know g^* and will update its beliefs and replan based on observed trajectory segments.

a) *Adversarial Evaluation Framework*: Rather than solving analytically for an equilibrium, we adopt an empirical adversarial evaluation framework. We implement Adversarial RRT* for Agent D and the IRL-based particle filter with game-theoretic MPC for Agent I, then evaluate their performance through simulation. Key performance metrics include:

- For Agent D: Observer classification accuracy at the time of goal completion, $P_\theta(g^* | \xi_D)$, and path length ratio $J_{\text{path}}(\xi_D^*)/J_{\text{path}}(\xi_D^{\text{opt}})$ where ξ_D^{opt} is the optimal (shortest) path
- For Agent I: Goal inference accuracy $\mathbb{1}[\hat{g}(T) = g^*]$ and time-to-intercept relative to Agent D's arrival time
- Overall: Success rate for each agent—does Agent D successfully reach g^* before being intercepted, or does Agent I successfully intercept Agent D?

b) *Interaction Dynamics*: The game proceeds as follows. Agent D generates a deceptive trajectory using Adversarial RRT*. Agent I observes the trajectory incrementally and updates its belief distribution over goals using the IRL-learned behavioral model and particle filtering. Based on the updated belief, Agent I replans its interception trajectory using game-theoretic MPC. This cycle continues until one of two termination conditions is met: (a) Agent D reaches the true goal g^* , or (b) Agent I successfully intercepts Agent D (i.e., $\|x_I(t) - x_D(t)\| < \epsilon_{\text{intercept}}$ for some small threshold $\epsilon_{\text{intercept}}$).

V. METHODOLOGY

A. Overview

The project is implemented in Python, chosen for its extensive ecosystem of machine learning and scientific computing libraries. In particular, we use JAX (via the Equinox library) as

the primary framework for differentiable programming, automatic differentiation, and GPU-accelerated computation.

The core of the project involves two competing agents in a continuous two-dimensional workspace. The first agent employs a deceptive motion planning algorithm building on Adversarial RRT* [1], which augments the sampling-based RRT* planner with a learned deception cost to generate trajectories that minimize an observer's ability to infer the agent's true goal. Entropy-based deceptive planning techniques [2], [5] inform the design of the deception objective. The second agent utilizes an identification and interception strategy, drawing on techniques from inverse reinforcement learning and game-theoretic prediction [3], [4], [8], with the objective of recognizing the deceptive agent's true goal and planning an intercept trajectory in real time. The approach of coupling adversary identification with reactive planning follows the framework proposed by Netter and Vamvoudakis [7], adapted to the adversarial evaluation setting. Both agents thus operate well beyond basic shortest-path planning: the deceptive agent layers a learned deception metric on top of RRT*, while the interceptor combines online goal inference with reactive replanning.

Of note, despite several of the referenced algorithms being originally developed for three-dimensional environments (e.g., first-person shooter game maps [4]), we work in a continuous two-dimensional workspace for ease of development and in the interest of completing the project within the semester. The simulation environment is custom-built in Python, providing full control over the experimental setup and enabling systematic evaluation of each algorithm's performance under varying conditions.

B. Implementation

Implementation is currently in progress, following a structured development approach. The first phase—comprehensive software specification—has been completed. This phase establishes the architectural foundation and detailed implementation plan before writing code, which is critical for a project of this complexity involving two interdependent agents, neural network components, and JAX-based differentiable programming.

1) *Software Specifications*: A complete set of implementation-ready software specifications has been developed and documented in `./docs/spec/`. The specifications follow an agent-based architectural pattern, organizing the codebase into four primary packages: `src/deceptive/` for the deceptive agent, `src/interceptor/` for the interceptor agent, `src/shared/` for common components (workspace, trajectories, collision detection), and `src/simulation/` for the simulation controller and evaluation framework.

The specifications include detailed pseudocode for all core algorithms. For the deceptive agent, this encompasses the

Adversarial RRT* planner with integrated deception cost evaluation, the RNN-based surrogate observer network architecture (implemented using Equinox), and the RRT* tree data structure. For the interceptor agent, specifications detail the maximum entropy IRL implementation for learning the deceptive agent’s behavioral model, the particle filter for maintaining and updating belief distributions over candidate goals, and the game-theoretic MPC formulation for computing interception controls.

All data formats and schemas have been fully specified. Training data formats for both the RNN observer (synthetic trajectory dataset with goal labels) and IRL module (expert demonstrations from the deceptive agent) are defined using HDF5 storage with documented array shapes and metadata. Complete YAML configuration schemas specify workspace geometry (bounds and obstacles), agent parameters (kinodynamic constraints, planner hyperparameters), and simulation settings. Output data formats cover trajectory storage (CSV and NumPy formats), performance metrics (observer accuracy, path length ratio, belief entropy), and visualization specifications (workspace plots, belief evolution charts).

JAX-specific implementation patterns are documented to ensure efficient GPU-accelerated execution. The specifications identify which components should be JIT-compiled (collision checking, cost functions, neural network inference) and which should not (RRT* tree construction with dynamic branching). Strategies for using `vmap` for batch parallelization (collision checking for multiple points, particle filter updates), `pytree` registration for custom data structures, and proper PRNG key management for reproducibility are all detailed.

Testing strategies have been defined across three levels. Unit tests will validate individual components such as collision detection algorithms (point-in-circle, point-in-polygon ray casting), trajectory interpolation accuracy, and RRT* tree operations. Integration tests will verify the complete deceptive planning pipeline (RRT* with observer network) and interception pipeline (particle filter with MPC). Validation tests will ensure algorithmic correctness: RRT* should converge to optimal paths when the deception weight $\alpha = 1$, the RNN observer should achieve greater than 80% classification accuracy on held-out test trajectories, and the particle filter should converge to the true goal given sufficient observations.

2) *Current Development Status:* With the architectural specifications complete, implementation is proceeding in the following order. The first development phase focuses on shared infrastructure: the 2D workspace representation with obstacle modeling, collision detection primitives, trajectory data structures and interpolation utilities, and kinodynamic constraint enforcement. These components form the foundation for both agents and will be implemented first to enable parallel development of agent-specific modules.

The second phase addresses neural network training. The RNN observer will be trained on synthetically generated optimal trajectories to each candidate goal, providing the deceptive agent with a surrogate observer for evaluating trajectory deceptiveness. Concurrently, the IRL module will be trained on demonstration data collected from the deceptive agent executing trajectories with various deception weights, enabling the interceptor to predict the deceiver’s likely future actions.

The third phase implements agent-specific planning algorithms. For the deceptive agent, this includes the Adversarial RRT* planner integrating path cost and deception cost, and the deception cost evaluator that queries the trained observer network. For the interceptor agent, this includes the particle filter for belief tracking and the game-theoretic MPC solver for computing interception controls.

The final phase integrates all components into the simulation controller, implements the adversarial game loop, and develops the metrics collection and visualization pipeline. This phase will also include comprehensive testing and validation against the criteria established in the specifications.

Currently, the project has just finished specification, and will soon begin development—starting first with the shared infrastructure components in `src/shared/`.

C. Procedure

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